Limits 1.1–1.2 · Cram sheet

Honors Precalculus BC · non-calculator quiz · limits from graphs, tables and algebra

The Method

substitute first	number: done. nonzero/0: DNE, done. 0/0: simplify, then substitute.
factor	polynomials: GCF, trinomial, $a^2 - b^2$, cubes; cancel the factor that makes 0.
conjugate	square roots: multiply top and bottom; $a^2 - b^2$, no FOIL.
LCD	fraction inside a fraction: multiply top and bottom by the LCD; distribute the negative.
notation	lim on every line until you substitute; never "= 0/0"; DNE stated about the limit.

Graphs

limit vs f(c)	limit = height the curve heads toward; f(c) = filled dot.
two-sided	left = right = L, or the limit does not exist.
all values of a	both sides reach the height; not one-sided spots, lone dots or open-interval ends.
two graphs	write the property, then read each limit from its own graph.
asymptote	both down: $-\infty$; both up: ∞; opposite: DNE.

The Rest

piecewise	at the break, left piece for c^-, right piece for c^+.
absolute value	$(x - a)/\lvert x - a\rvert$ is 1 right of a, −1 left of a.
tables	closest values each side, round to thousandths; an estimate, not proof.
difference quotient	build f(x + h), subtract, cancel h, substitute 0 for h.
average rate of change	[f(b) − f(a)]/(b − a), same order top and bottom.

Formulas

limit of a constant	$\lim_{x \to c} A = A$
limit of x	$\lim_{x \to c} x = c$ *(only for y = x)*
sum and difference	$\lim [f(x) \pm g(x)] = \lim f(x) \pm \lim g(x)$
product	$\lim [f(x) \cdot g(x)] = \lim f(x) \cdot \lim g(x)$
constant multiple	$\lim [k\, g(x)] = k \lim g(x)$ *(k is a number, no variable)*
power	$\lim [f(x)]^n = [\lim f(x)]^n$
root	$\lim \sqrt[n]{f(x)} = \sqrt[n]{\lim f(x)}$ *(f(x) ≥ 0 if n is even)*
quotient	$\lim f(x)/g(x) = \lim f(x) / \lim g(x)$ *(only if lim g(x) ≠ 0)*
polynomial	$\lim_{x \to c} P(x) = P(c)$
two-sided limit	$\lim_{x \to c} f(x) = L$ only if $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = L$
average rate of change	$[f(b) - f(a)]/(b - a)$
difference quotient	$[f(x + h) - f(x)]/h,\ h \neq 0$
difference of squares	$a^2 - b^2 = (a - b)(a + b)$ *(x − 9 = (√x − 3)(√x + 3))*
sum and difference of cubes	$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$

Watch Out

- $4 - (x + 4) = -x$: distribute the negative to both terms.
- Do not FOIL a conjugate product: $(a - b)(a + b) = a^2 - b^2$.
- $h \to 0$: substitute 0 for h, not x.
- Two-graph questions: write the property before the numbers.
- An isolated dot or a one-sided approach never counts for "all values of a".

Key: DNE = does not exist · GCF = greatest common factor · LCD = least common denominator · FOIL = first, outer, inner, last